

Adjacency eigenvalues do not determine homomorphism counts

Conjecture 00000008905 ziangni-sys

The source claims reconstruction of homomorphism counts from eigenvalues. Under the standard graph-homomorphism meaning of this statement, even the complete adjacency spectrum with algebraic multiplicities is insufficient. Both examples below have the same number of vertices; no loss of isolated vertices or of spectral multiplicities is involved.

Counterexample. *There are simple graphs G, H on six vertices with the same adjacency characteristic polynomial but*

$$\text{hom}(P_3, G) = 20 \neq 16 = \text{hom}(P_3, H),$$

where P_3 is the path on three vertices. Thus no function of the adjacency eigenvalue multiset reconstructs these homomorphism counts.

Proof. Take $G = K_{1,4} \sqcup K_1$ and $H = C_4 \sqcup 2K_1$. Explicitly, use vertex set $\{0, 1, 2, 3, 4, 5\}$ for both. In G , the edges join 0 to each of 1, 2, 3, 4, and 5 is isolated. In H , all four edges join the part $\{0, 2\}$ to the part $\{1, 3\}$, and 4, 5 are isolated. All edges are undirected, with no loops.

For the star, the space spanned by the central coordinate and the sum of its four leaf coordinates has adjacency matrix

$$\begin{pmatrix} 0 & 4 \\ 1 & 0 \end{pmatrix}.$$

The leaf-coordinate subspace of sum zero has dimension three and eigenvalue zero. Adding the isolated vertex gives

$$\chi_G(t) = t^4(t^2 - 4).$$

For $C_4 = K_{2,2}$, the two constant-on-part vectors span a space with adjacency matrix

$$\begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}.$$

The two within-part differences have eigenvalue zero; the two isolated vertices add two more zero eigenvalues. Hence

$$\chi_H(t) = t^4(t^2 - 4) = \chi_G(t).$$

Thus both complex eigenvalue multisets are $\{2, -2, 0, 0, 0, 0\}$.

For any simple graph J , a homomorphism from P_3 is determined by the image v of its middle vertex and the ordered images of its two endpoints. Each endpoint independently has $\deg_J(v)$ choices. Repeated endpoint images are allowed in a graph homomorphism. Therefore

$$\text{hom}(P_3, J) = \sum_{v \in V(J)} \deg_J(v)^2.$$

The degree lists are $(4, 1, 1, 1, 1, 0)$ and $(2, 2, 2, 2, 0, 0)$, respectively, giving 20 and 16. Any reconstruction function applied to their common eigenvalue multiset would have to return both numbers, an impossibility. $\square$

Scope and formal verification. Only the asserted spectral reconstruction of unrestricted graph-homomorphism counts is refuted. This does not deny that adjacency spectra determine closed-walk counts. The Lean project constructs actual `SimpleGraph` objects and their adjacency matrices, proves equality of their characteristic polynomials over $\mathbb{Q}$ using an explicit invertible intertwiner, transports equality to $\mathbb{C}$, and proves equality of the full root multisets. It counts the actual types of `SimpleGraph.Hom` maps through an explicit equivalence with the complete finite set of adjacency-preserving functions. The final theorem rules out every reconstruction function on eigenvalue multisets. Finite computations use kernel-checked `decide`, never `native_decide`. Run `lake build in lean/`; the project pins Lean 4.19.0 and Mathlib in its manifest.